DIMS Letters API — Architecture

A layered walkthrough of the system, scaling from a 30-second overview for non-technical readers down to a component-level view for engineers. Every Mermaid block below renders as an image directly on GitHub.

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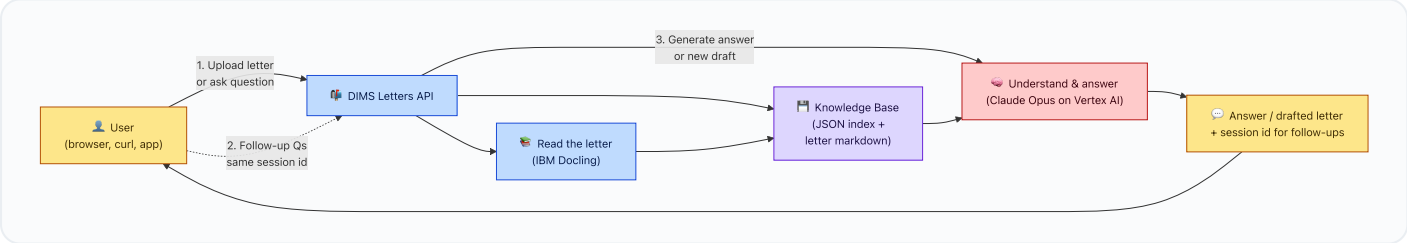
1. The 30-second summary

> **DIMS Letters API** lets a user **upload an FDA letter (PDF / DOCX) and chat with it in plain English** — or ask the system to **draft a new letter** in Dr. Reddy's house style based on past letters. It runs on FastAPI, uses **IBM Docling** to read PDFs, and **Anthropic Claude Opus on Google Vertex AI** to understand and write text. No database — just a JSON index file plus the extracted markdown on disk.



Read it like a story:

1. User uploads a letter or asks a question.
2. The API reads the PDF (Docling) and stores the text in a small knowledge base.
3. Claude Opus answers the question — or drafts a brand-new letter — using past letters as examples.
4. The user gets the answer and can keep asking follow-ups via a session id.

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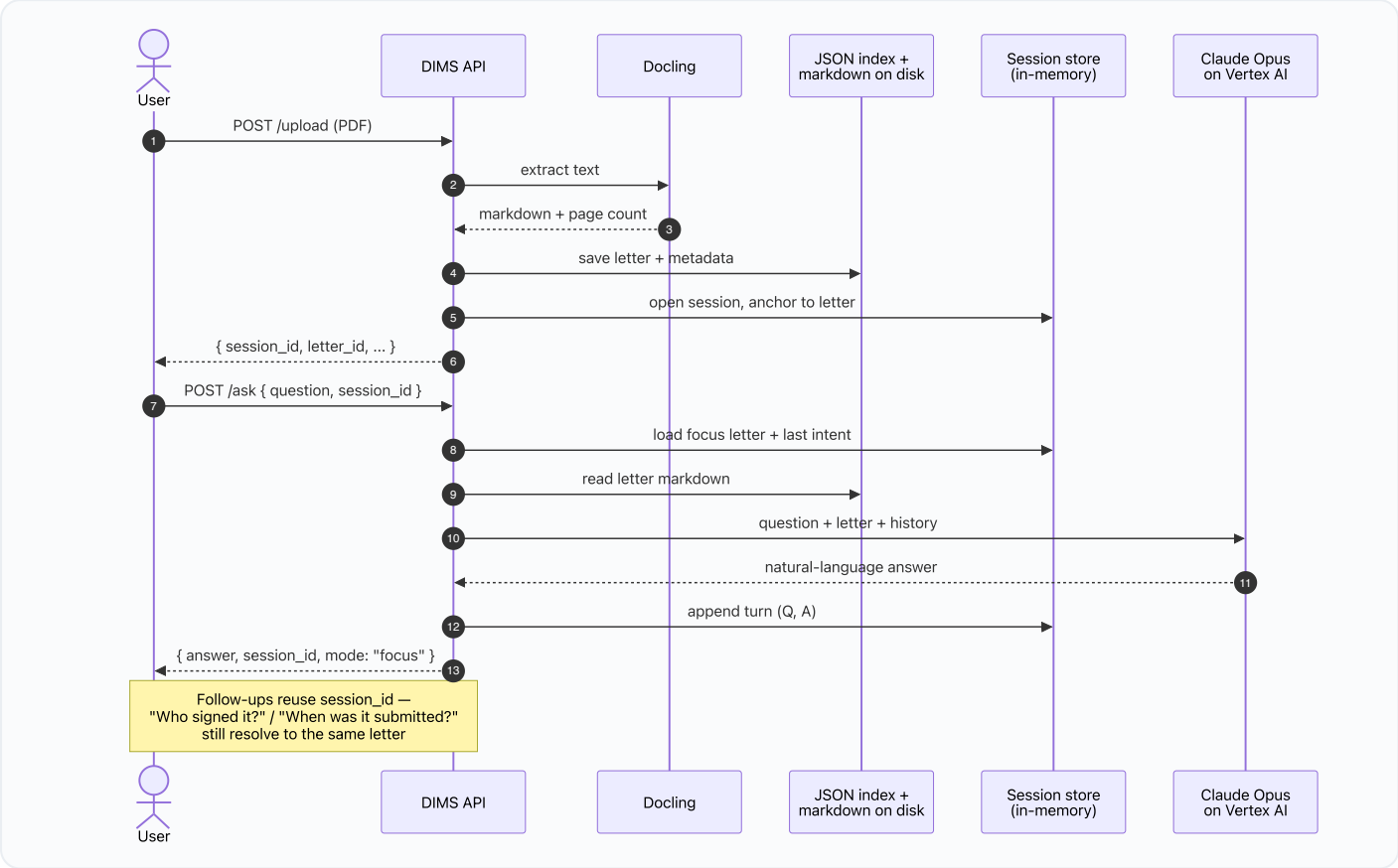
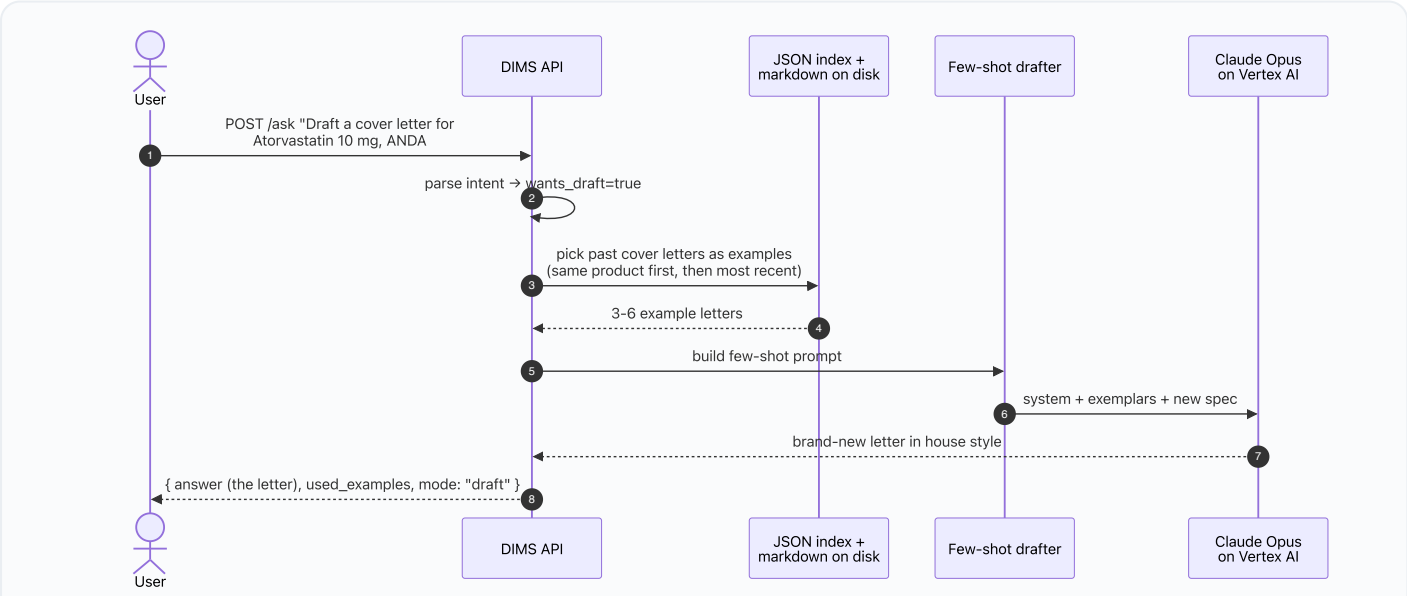
2. Two user journeys (step-by-step)

Journey A — *"I have a letter. Let me ask questions about it."*

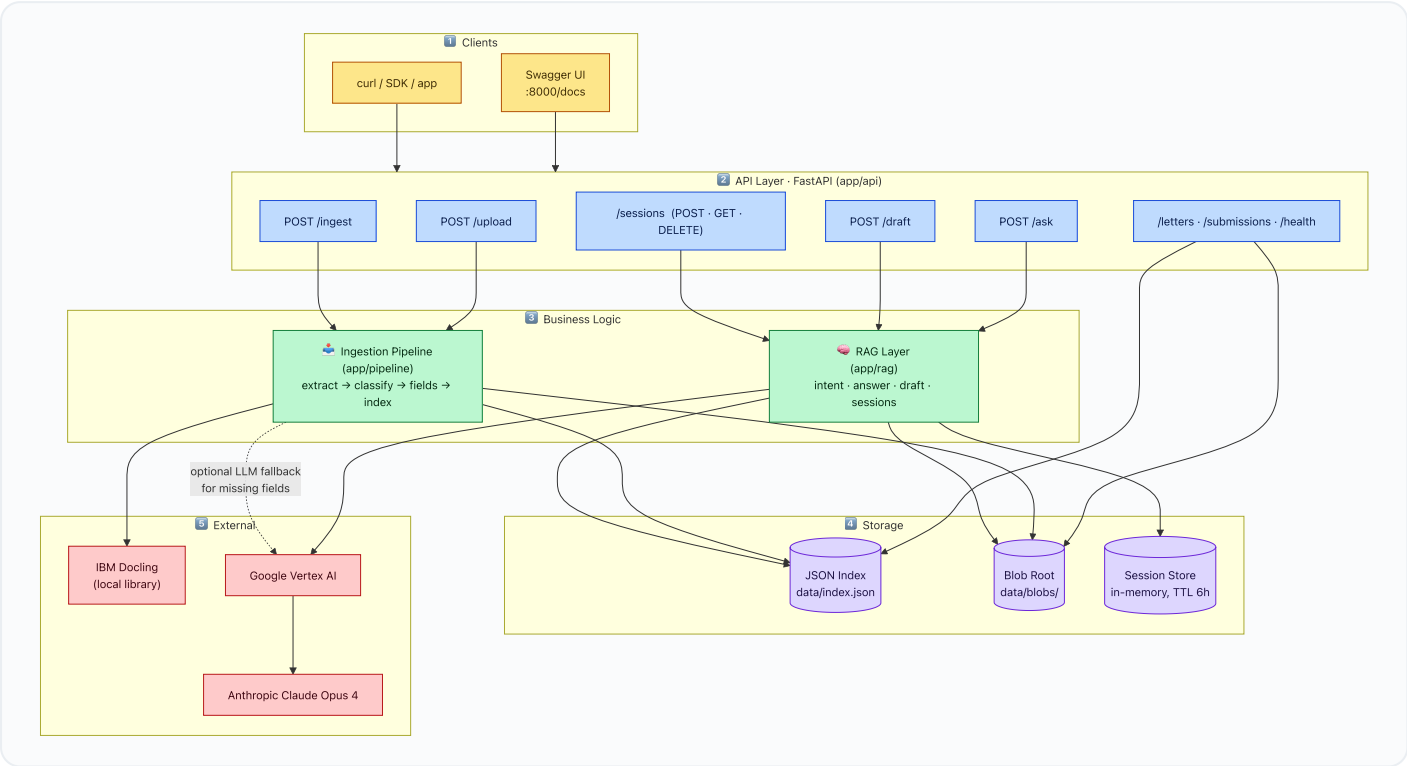


Journey B — *"I don't have the letter yet — draft one for me."*



3. Technical architecture (for engineers)

Five clear layers, top to bottom. Each box has only its role; the table after the diagram maps boxes to actual files.



File map (what lives where)

Layer	Component	Code location	One-line role
API	Routes	`app/api/routes.py`	All HTTP endpoints
API	Schemas	`app/api/schemas.py`	Request / response Pydantic models
API	App entry	`app/main.py`	FastAPI app; OpenAPI 3.0 downgrade so Swagger renders the upload picker
Pipeline	Adapters	`app/adapters/`	Pluggable inputs (file-store, RDB stub)
Pipeline	Orchestrator	`app/pipeline/ingest.py`	Folder ingest + direct upload
Pipeline	Extract	`app/pipeline/extract.py`	Docling PDF → markdown + page map
Pipeline	Fields	`app/pipeline/fields.py`	Classify letter kind, pull header fields (regex + LLM fallback)
RAG	Intent parser	`app/rag/intent.py`	Question → `{product, kind, anda, seq, wants_draft}`
RAG	Answer	`app/rag/answer.py`	Lookup / focus / disambiguate / draft fall-through
RAG	Drafter	`app/rag/draft.py`	Few-shot prompt build, exemplar selection
RAG	Sessions	`app/rag/sessions.py`	Multi-turn memory (focus letter, intent carry-over, history)
Storage	Index	`app/index.py` → `data/index.json`	All submission + letter records

Layer	Component	Code location	One-line role
Storage	Blobs	`data/blobs/originals/`, `data/blobs/markdown/`	Original files + Docling output
LLM	Vertex client	`app/llm.py`	`AnthropicVertex` (Claude Opus 4 on Vertex)
Config	Settings	`app/config.py` + `.env`	All env-driven knobs

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4. The modes `/ask` can return (cheat sheet)

`mode`	When	What you get back
`lookup`	Question matched exactly one indexed letter	Claude's summary + the matched letter
`focus`	Same session asked a follow-up about its anchored letter	Answer against the focused letter
`disambiguate`	Multiple letters match (e.g. *"everything for Voclosporin"*)	List of candidates; ask the user to narrow down
`draft`	User said "draft" / no letter exists for that product	A newly drafted letter in house style
`not_found`	Not enough info to look up *or* draft	Friendly error explaining what's missing

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5. Configuration knobs

All driven by environment variables (or a `.env` file at repo root):

Variable	Purpose
`SOURCES__FILESTORE__ROOT`	Folder root for batch ingest (`POST /ingest`)
`BLOB_ROOT`	Where Docling output + originals are persisted (default `data/blobs`)
`INDEX_PATH`	Path to the JSON index (default `data/index.json`)
`VERTEX_PROJECT_ID`	GCP project hosting Vertex AI
`VERTEX_REGION`	Vertex region (default `us-east5`)
`ANTHROPIC_MODEL`	Claude model id (default `claude-opus-4@20250514`)
`ANTHROPIC_MAX_TOKENS`	Per-call cap (default 4096)
`GOOGLE_APPLICATION_CREDENTIALS`	Path to GCP service-account JSON (ADC)
`LOG_LEVEL`	Python logging level (default `INFO`)

If `VERTEX_PROJECT_ID` is unset, the API still works for ingest, upload, listing, and verbatim letter retrieval — only the LLM-powered answers and drafts degrade gracefully with a clear "LLM not configured" message.